Example 5.32

Find $\frac{d}{dx}(e^{3x})$ from first principle.

Solution

Let $f(x) = e^{3x}$

$\therefore f(x+h) = e^{3(x+h)}$

$$\frac{d}{dx}(f(x)) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{e^{3(x+h)}-e^{3x}}{h}$$

$$= \lim_{h\to 0}\frac{e^{3x}\cdot e^{3h}-e^{3x}}{h}$$

$$= \lim_{h\to 0}\frac{e^{3x}(e^{3h}-1)}{h}$$

$$= 3e^{3x}\lim_{h\to 0}\frac{e^{3h}-1}{3h}$$

$$= 3e^{3x}\times 1 \quad [\because \lim_{x\to 0}\frac{e^x-1}{x} = 1]$$

$$= 3e^{3x}$$

Exercise 5.4

1. Find the derivative of the following functions from first principle.

   (i) x^2 (ii) e^{-x} (iii) $\log(x+1)$

5.3 Differentiation Techniques

In this section we will discuss about different techniques to obtain the derivatives of the given functions.

5.3.1 Some standard results [formulae]

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}(x) = 1$
3. $\frac{d}{dx}(k) = 0$, k is a constant
4. $\frac{d}{dx}(kx) = k$, k is a constant
5. $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$
6. $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$
7. $\frac{d}{dx}(e^x) = e^x$
8. $\frac{d}{dx}(e^{ax}) = ae^{ax}$
9. $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$
10. $\frac{d}{dx}(e^{-x^2}) = -2x\,e^{-x^2}$
11. $\frac{d}{dx}(a^x) = a^x \log_e a$, $a > 0$
12. $\frac{d}{dx}(\log_e x) = \frac{1}{x}$
13. $\frac{d}{dx}\log_e(x+a) = \frac{1}{x+a}$
14. $\frac{d}{dx}(\log_e(ax+b)) = \frac{a}{(ax+b)}$
15. $\frac{d}{dx}(\log_a x) = \frac{1}{x\log_e a}$, a>0 and a≠1
16. $\frac{d}{dx}(\sin x) = \cos x$
17. $\frac{d}{dx}(\cos x) = -\sin x$
18. $\frac{d}{dx}(\tan x) = \sec^2 x$
19. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
20. $\frac{d}{dx}(\sec x) = \sec x \tan x$
21. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

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